SECTION 1: DOUBLE RIVETED LAP JOINT UNDER TENSION, SHEAR, AND CRUSHING STRESSES

System Parameters: Plate Thickness $t = 15\ \text{mm}$, Rivet Diameter $d = 25\ \text{mm}$, Pitch $p = 75\ \text{mm}$, Ultimate Tensile Stress $\sigma_\text{u} = 400\ \text{MPa}$, Ultimate Shear Stress $\tau_\text{u} = 320\ \text{MPa}$, Ultimate Crushing Stress $\sigma_\text{cu} = 640\ \text{MPa}$, Factor of Safety $\text{FOS} = 4$.

Design Protocol: Evaluate plate tearing, rivet shearing ($n = 2$), and rivet crushing strengths to determine minimum rupturing force P_u per pitch; compute working load $P = P_\text{u}$ / FOS and evaluate working stresses.

1. Tearing Strength of Plate (P_t)
Plate tearing capacity per pitch length

$$P_t = (p - d)t \cdot \sigma_\text{u}$$
$$= (75 - 25) \times 15 \times 400$$
$$= 300000\ \text{N}$$
$$\Rightarrow P_t = 300\ \text{kN}$$

2. Shearing Strength of Rivets (P_s)
Single shear capacity of 2 rivets per pitch length

$$P_s = n \cdot \frac{\pi}{4} d^2 \cdot \tau_\text{u}$$
$$= 2 \times \frac{\pi}{4}(25)^2 \times 320$$
$$= 314160\ \text{N}$$
$$\Rightarrow P_s = 314.16\ \text{kN}$$

3. Crushing Strength of Rivets (P_c)
Bearing capacity of 2 rivets per pitch length

$$P_c = n \cdot d \cdot t \cdot \sigma_\text{cu}$$
$$= 2 \times 25 \times 15 \times 640$$
$$= 480000\ \text{N}$$
$$\Rightarrow P_c = 480\ \text{kN}$$

4. Rupturing Force per Pitch (P_u)
Minimum strength governing joint rupture

$$P_u = \min(P_t, P_s, P_c)$$
$$= \min(300, 314.16, 480)$$
$$\Rightarrow P_u = 300\ \text{kN}$$

5. Safe Working Load (P)
Applied force per pitch at FOS = 4

$$P = \frac{P_u}{\text{FOS}}$$
$$= \frac{300}{4}$$
$$= 75\ \text{kN}$$
$$= 75000\ \text{N}$$

6. Actual Tensile Stress (σ_t)
Induced tensile stress on main plate

$$\sigma_t = \frac{P}{(p - d)t}$$
$$= \frac{75000}{(75 - 25) \times 15}$$
$$= 100\ \text{MPa}$$

7. Actual Shear Stress (τ)

Induced shear stress in rivets

$$\tau = \frac{P}{n \cdot \frac{\pi}{4} d^2}$$
$$= \frac{75000}{2 \times \frac{\pi}{4}(25)^2}$$
$$= 76.4 \text{ MPa}$$

8. Actual Crushing Stress (σ_c)

Induced crushing stress on rivets

$$\sigma_c = \frac{P}{n \cdot d \cdot t}$$
$$= \frac{75000}{2 \times 25 \times 15}$$
$$= 100 \text{ MPa}$$

Design Output

Rupturing Force Pu = 300 kN | Working Load P = 75 kN | Working Stresses: sigma_t = 100 MPa, tau = 76.4 MPa, sigma_c = 100 MPa

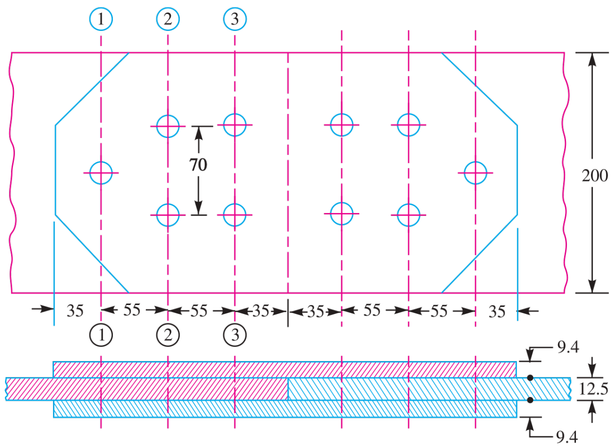


Fig. 9.20. All dimensions in mm.

Fig. 9.20: Double Riveted Lap Joint Scheme

SECTION 2: EFFICIENCY OF SINGLE AND DOUBLE RIVETED LAP JOINTS

System Parameters: Plate Thickness $t = 6$ mm, Rivet Diameter $d = 20$ mm, Case 1 Pitch $p_1 = 50$ mm ($n = 1$), Case 2 Pitch $p_2 = 65$ mm ($n = 2$), Allowable Stresses: Tensile $\sigma_t = 120$ MPa, Shear $\tau = 90$ MPa, Crushing $\sigma_c = 180$ MPa.

Design Protocol: Calculate un-punched solid plate strength, tearing, shearing, and crushing strengths for single and double riveted lap joints; compute joint efficiencies.

---

1. Solid Plate Strength — Single Lap (P)
Strength of un-punched plate ($p = 50$ mm)

$$P = p \cdot t \cdot \sigma_t$$
$$= 50 \times 6 \times 120$$
$$= 36000 \text{ N}$$

2. Tearing Strength — Single Lap (P_t)
Net section tensile capacity

$$P_t = (p - d)t \cdot \sigma_t$$
$$= (50 - 20) \times 6 \times 120$$
$$= 21600 \text{ N}$$

3. Shearing Strength — Single Lap (P_s)
Single shear capacity ($n = 1$)

$$P_s = n \cdot \frac{\pi}{4} d^2 \cdot \tau$$
$$= 1 \times \frac{\pi}{4}(20)^2 \times 90$$
$$= 28274 \text{ N}$$

4. Crushing Strength — Single Lap (P_c)
Bearing capacity ($n = 1$)

$$P_c = n \cdot d \cdot t \cdot \sigma_c$$
$$= 1 \times 20 \times 6 \times 180$$
$$= 21600 \text{ N}$$

5. Efficiency — Single Riveted Lap Joint (η_1)
Ratio of minimum strength to solid strength

$$\eta_1 = \frac{\min(P_t, P_s, P_c)}{P} \times 100\%$$
$$= \frac{21600}{36000} \times 100\%$$
$$\Rightarrow \boldsymbol{\eta_1 = 60.0\%}$$

6. Solid Plate Strength — Double Lap (P)
Strength of un-punched plate ($p = 65$ mm)

$$P = p \cdot t \cdot \sigma_t$$
$$= 65 \times 6 \times 120$$
$$= 46800 \text{ N}$$

7. Tearing Strength — Double Lap (P_t)
Net section tensile capacity

$$P_t = (p - d)t \cdot \sigma_t$$
$$= (65 - 20) \times 6 \times 120$$
$$= 32400 \text{ N}$$

8. Shearing Strength — Double Lap (P_s)
Single shear capacity ($n = 2$)

$$P_s = n \cdot \frac{\pi}{4} d^2 \cdot \tau$$
$$= 2 \times \frac{\pi}{4}(20)^2 \times 90$$
$$= 56548 \text{ N}$$

9. Crushing Strength — Double Lap (P_c)
Bearing capacity ($n = 2$)

$$P_c = n \cdot d \cdot t \cdot \sigma_c$$
$$= 2 \times 20 \times 6 \times 180$$
$$= 43200 \text{ N}$$

10. Efficiency — Double Riveted Lap Joint (η_2)
Ratio of minimum strength to solid strength

$$\eta_2 = \frac{\min(P_t, P_s, P_c)}{P} \times 100\%$$
$$= \frac{32400}{46800} \times 100\%$$
$$\Rightarrow \boldsymbol{\eta_2 = 69.2\%}$$

Design Output

Single Riveted Lap Efficiency eta1 = 60.0% | Double Riveted Lap Efficiency eta2 = 69.2%

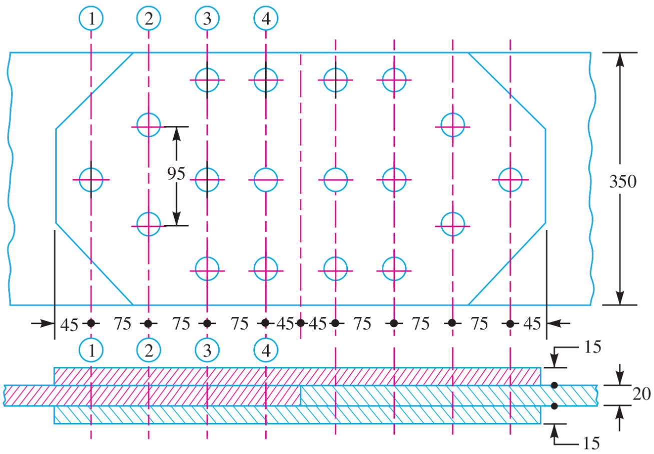


Fig. 9.21. All dimensions in mm.

Fig. 9.21: Single Riveted Lap Joint Scheme

SECTION 3: EFFICIENCY OF DOUBLE RIVETED DOUBLE COVER BUTT JOINT

System Parameters: Plate Thickness $t = 20\text{ mm}$, Rivet Diameter $d = 25\text{ mm}$, Pitch $p = 100\text{ mm}$, Allowable Stresses: Tensile $\sigma_t = 120\text{ MPa}$, Shear $\tau = 100\text{ MPa}$, Crushing $\sigma_c = 150\text{ MPa}$, Double Shear Factor = 2.0.

Design Protocol: Evaluate solid plate strength, plate tearing strength, rivet double shear strength ($n = 2$), and rivet crushing strength to determine joint efficiency.

1. Solid Plate Strength (P)
Un-punched solid plate tensile capacity

$$P = p \cdot t \cdot \sigma_t$$
$$= 100 \times 20 \times 120$$
$$= 240000\text{ N}$$

2. Tearing Strength of Plate (P_t)
Net section tensile strength per pitch

$$P_t = (p - d)t \cdot \sigma_t$$
$$= (100 - 25) \times 20 \times 120$$
$$= 180000\text{ N}$$

3. Shearing Strength of Rivets (P_s)
Double shear capacity of 2 rivets per pitch ($n = 2$, factor = 2)

$$P_s = 2 \cdot \left(2 \cdot \frac{\pi}{4} d^2\right) \cdot \tau$$
$$= 2 \times \left(2 \times \frac{\pi}{4}(25)^2\right) \times 100$$
$$= 196350\text{ N}$$

4. Crushing Strength of Rivets (P_c)
Bearing capacity of 2 rivets per pitch ($n = 2$)

$$P_c = n \cdot d \cdot t \cdot \sigma_c$$
$$= 2 \times 25 \times 20 \times 150$$
$$= 150000\text{ N}$$

5. Joint Efficiency (η)
Ratio of minimum strength to un-punched solid plate strength

$$\eta = \frac{\min(P_t, P_s, P_c)}{P} \times 100\%$$
$$= \frac{150000}{240000} \times 100\%$$
$$\Rightarrow \boldsymbol{\eta = 62.5\%}$$

Design Output

Double Cover Butt Joint Efficiency eta = 62.5% (Governed by Rivet Crushing)

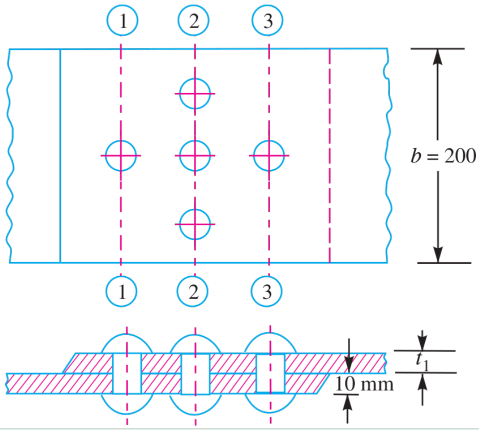


Fig. 9.22

Fig. 9.22: Double Riveted Double Cover Butt Joint

SECTION 4: DOUBLE RIVETED LAP JOINT — ZIG-ZAG RIVETING

System Parameters: Plate Thickness $t = 13$ mm, Tensile Stress $\sigma_t = 80$ MPa, Shear Stress $\tau = 60$ MPa, Crushing Stress $\sigma_c = 120$ MPa, Rivet Rows $n = 2$ (zig-zag).

Design Protocol: Calculate rivet hole diameter d via Unwin's formula, determine pitch p by equating $P_t = P_s$ and verify $p \leq p_{\max}$, compute row pitch p_b and margin m, evaluate governing failure mode ($\min(P_t, P_s, P_c)$), and calculate joint efficiency η.

---

1. Diameter of Rivet Hole (d)

Unwin's formula for $t = 13$ mm > 8 mm and IS standard size

$$d = 6\sqrt{t}$$
$$= 6\sqrt{13} = 21.63 \text{ mm}$$
$$\Rightarrow \textbf{Standard Hole Diameter } \boldsymbol{d = 23} \textbf{ mm}$$
$$\Rightarrow \textbf{Nominal Rivet Diameter } \boldsymbol{d_0 = 22} \textbf{ mm}$$

2. Tearing Resistance of Plate (P_t)

Net tensile area capacity per pitch length

$$P_t = (p - d)t \cdot \sigma_t$$
$$= (p - 23) \times 13 \times 80$$
$$= 1040(p - 23) \text{ N}$$

3. Shearing Resistance of Rivets (P_s)

Single shear capacity for 2 rivets per pitch ($n = 2$)

$$P_s = n \cdot \frac{\pi}{4} d^2 \cdot \tau$$
$$= 2 \times \frac{\pi}{4}(23)^2 \times 60$$
$$= 2 \times 415.48 \times 60$$
$$= \mathbf{49864 \text{ N}}$$

4. Pitch & Max Pitch Limit ($p, p_{\max}$)

Equating $P_t = P_s$ and enforcing I.B.R. maximum pitch limit

$$1040(p - 23) = 49864 \text{ N}$$
$$p - 23 = \frac{49864}{1040} = 47.95 \Rightarrow p = 70.95 \text{ mm}$$
$$p_{\max} = C \cdot t + 41.28 \text{ mm}$$
$$= 2.62(13) + 41.28 = 75.34 \text{ mm}$$
$$\Rightarrow \textbf{Adopt Pitch } \boldsymbol{p = 71} \textbf{ mm}$$

5. Distance Between Rows of Rivets (p_b)

Transverse row pitch for zig-zag riveting

$$p_b = 0.33p + 0.67d$$
$$= 0.33(71) + 0.67(23)$$
$$= 23.43 + 15.41 = 38.84 \text{ mm}$$
$$\Rightarrow \textbf{Adopt Row Pitch } \boldsymbol{p_b = 40} \textbf{ mm}$$

6. Margin (m)

Distance from rivet hole center to plate edge

$$m = 1.5d$$
$$= 1.5(23) = 34.5 \text{ mm}$$
$$\Rightarrow \textbf{Adopt Margin } \boldsymbol{m = 35} \textbf{ mm}$$

7. Failure Mode Analysis

Tearing (P_t), shearing (P_s), and crushing (P_c) capacities for adopted $p = 71$ mm

$$P_t = (p - d)t \cdot \sigma_t = (71 - 23)(13)(80) = 49920 \text{ N}$$

$$P_s = n \cdot \frac{\pi}{4} d^2 \cdot \tau = 2\left(\frac{\pi}{4}\right)(23)^2(60) = 49864 \text{ N (Go}$$

$$P_c = n \cdot d \cdot t \cdot \sigma_c = 2(23)(13)(120) = 71760 \text{ N}$$

$\Rightarrow$ **Governing Strength $P_u = P_s = 49864$ N**

$\Rightarrow$ **Failure Mode: Shearing of Rivets**

8. Solid Plate Strength & Joint Efficiency (P, η)

Un-punched solid plate strength and overall joint efficiency

$$P = p \cdot t \cdot \sigma_t$$

$$= 71 \times 13 \times 80$$

$$= 73840 \text{ N}$$

$$\eta = \frac{P_s}{P} \times 100\%$$

$$= \frac{49864}{73840} \times 100\%$$

$$\Rightarrow \eta = \mathbf{67.5\%}$$

Design Output Summary

Rivet Hole d = 23 mm | Pitch p = 71 mm | Row Pitch pb = 40 mm | Margin m = 35 mm | Failure Mode: Shearing | Efficiency eta = 67.5%

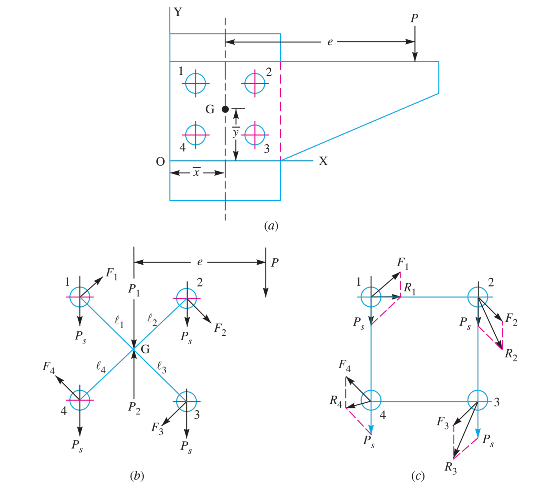


Fig. 9.23. Eccentric loaded riveted joint.

Fig. 9.23: Double Riveted Lap Joint with Zig-Zag Riveting

SECTION 5: TRIPLE RIVETED LAP JOINT — ZIG-ZAG PATTERN

System Parameters: Plate Thickness $t = 7$ mm, Tensile Stress $\sigma_t = 90$ MPa, Shear Stress $\tau = 60$ MPa, Crushing Stress $\sigma_c = 120$ MPa, Rivet Rows $n = 3$ (zig-zag).

Design Protocol: For thin plates ($t < 8$ mm), equate shearing strength to crushing strength ($P_s = P_c$) to determine rivet hole diameter d, equate tearing strength to shearing strength ($P_t = P_s$) to solve for pitch p, limit pitch by I.B.R. maximum pitch ($p \leq p_{\max}$), calculate row pitch p_b, and evaluate governing failure mode.

---

1. Shearing & Crushing Expressions (P_s, P_c)
Symbolic capacities for $t = 7$ mm < 8 mm ($n = 3$ rivets per pitch)

$$P_s = n \cdot \frac{\pi}{4} d^2 \cdot \tau = 3 \times \frac{\pi}{4} d^2 \times 60 = 141.4 d^2 \text{ N}$$

$$P_c = n \cdot d \cdot t \cdot \sigma_c = 3 \times d \times 7 \times 120 = 2520 d \text{ N}$$

2. Diameter of Rivet Hole (d)
Equating shearing strength P_s to crushing strength P_c

$$141.4 d^2 = 2520 d$$

$$d = \frac{2520}{141.4} = 17.8 \text{ mm}$$

$$\Rightarrow \textbf{Standard Hole Diameter} \;\; d = 19 \text{ mm}$$

$$\Rightarrow \textbf{Nominal Rivet Diameter} \;\; d_0 = 18 \text{ mm}$$

3. Tearing & Shearing Resistances (P_t, P_s)
Capacities evaluated for standard hole diameter $d = 19$ mm

$$P_t = (p - d) t \cdot \sigma_t = (p - 19) \times 7 \times 90 = 630(p - 1$$

$$P_s = 141.4 d^2 = 141.4(19)^2 = 51045 \text{ N}$$

4. Rivet Pitch & Max Pitch Limit ($p, p_{\max}$)
Equating $P_t = P_s$ and enforcing I.B.R. maximum pitch limit

$$630(p - 19) = 51045 \text{ N}$$

$$p - 19 = \frac{51045}{630} = 81 \Rightarrow p = 100 \text{ mm}$$

$$p_{\max} = C \cdot t + 41.28 \text{ mm}$$

$$= 3.47(7) + 41.28 = 65.57 \text{ mm}$$

$$\Rightarrow \textbf{Adopt Pitch} \;\; p = p_{\max} = 66 \text{ mm}$$

5. Distance Between Rows of Rivets (p_b)
Transverse row pitch for zig-zag riveting

$$p_b = 0.33 p + 0.67 d$$

$$= 0.33(66) + 0.67(19)$$

$$= 21.78 + 12.73 = 34.51 \text{ mm}$$

$$\Rightarrow \textbf{Adopt Row Pitch} \;\; p_b = 34.5 \text{ mm}$$

6. Mode of Failure Analysis
Tearing (P_t), shearing (P_s), and crushing (P_c) capacities for adopted $p = 66$ mm

$$P_t = (p - d) t \cdot \sigma_t = (66 - 19)(7)(90) = 29610 \text{ N } ($$

$$P_s = n \cdot \frac{\pi}{4} d^2 \cdot \tau = 3\left(\frac{\pi}{4}\right)(19)^2(60) = 51045 \text{ N}$$

$$P_c = n \cdot d \cdot t \cdot \sigma_c = 3(19)(7)(120) = 47880 \text{ N}$$

$$\Rightarrow \textbf{Governing Strength} \;\; P_u = P_t = 29610 \text{ N}$$

$$\Rightarrow \textbf{Failure Mode: Tearing off the Plate}$$

Design Output Summary

Rivet Hole d = 19 mm | Adopted Pitch p = 66 mm | Row Pitch pb = 34.5 mm | Failure Mode: Tearing of Plate

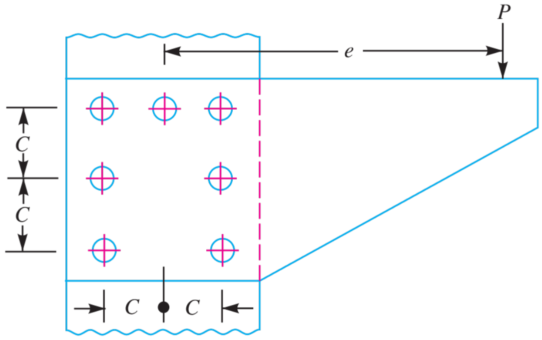


Fig. 9.24

Fig. 9.24: Triple Riveted Lap Joint with Zig-Zag Pattern

SECTION 6: SINGLE RIVETED DOUBLE STRAP BUTT JOINT

System Parameters: Main Plate Thickness $t = 10$ mm, Tensile Stress $\sigma_t = 80$ MPa, Shear Stress $\tau = 60$ MPa, Double Shear Factor = 1.875, Number of Rivets $n = 1$.

Design Protocol: Calculate rivet hole diameter d via Unwin's formula, determine double shear capacity P_s, state symbolic tearing strength (P_t), solve for pitch p, limit by I.B.R. maximum pitch ($p \leq p_{\max}$), compute cover strap thickness $t_1 = 0.625t$, evaluate governing failure mode, and calculate joint efficiency η.

---

1. Diameter of Rivet Hole (d)
Unwin's formula for $t = 10$ mm > 8 mm and IS standard size

$$d = 6\sqrt{t}$$
$$= 6\sqrt{10} = 18.97 \text{ mm}$$
$$\Rightarrow \textbf{Standard Hole Diameter } \ d = 19 \text{ mm}$$
$$\Rightarrow \textbf{Nominal Rivet Diameter } \ d_0 = 18 \text{ mm}$$

2. Tearing Resistance of Plate (P_t)
Net tensile area capacity per pitch length

$$P_t = (p - d)t \cdot \sigma_t$$
$$= (p - 19) \times 10 \times 80$$
$$= 800(p - 19) \text{ N}$$

3. Double Shear Strength of Rivets (P_s)
Double shear capacity for single rivet ($n = 1$, factor 1.875)

$$P_s = n \cdot 1.875 \cdot \frac{\pi}{4} d^2 \cdot \tau$$
$$= 1 \times 1.875 \times \frac{\pi}{4}(19)^2 \times 60$$
$$= 1.875 \times 283.53 \times 60$$
$$= \textbf{31900 N}$$

4. Pitch & Max Pitch Limit ($p, p_{\max}$)
Equating $P_t = P_s$ and enforcing I.B.R. maximum pitch limit

$$800(p - 19) = 31900 \text{ N}$$
$$p - 19 = \frac{31900}{800} = 39.87 \Rightarrow p = 58.87 \text{ mm}$$
$$p_{\max} = C \cdot t + 41.28 \text{ mm}$$
$$= 1.75(10) + 41.28 = 58.78 \text{ mm}$$
$$\Rightarrow \textbf{Adopt Pitch } \ p = p_{\max} = 60 \text{ mm}$$

5. Cover Strap Thickness (t_1)
Standard proportion for equal width double cover straps

$$t_1 = 0.625t$$
$$= 0.625(10)$$
$$\Rightarrow \textbf{Strap Thickness } \ t_1 = 6.25 \text{ mm}$$

6. Failure Mode & Joint Efficiency Verification (η)

Tearing (P_t), shearing (P_s), solid plate strength (P), and efficiency

$$P_t = (p-d)t \cdot \sigma_t = (60-19)(10)(80) = 32800 \text{ N}$$

$$P_s = 31900 \text{ N} \Rightarrow \textbf{Failure Mode: Shearing of R}$$

$$P = p \cdot t \cdot \sigma_t = 60(10)(80) = 48000 \text{ N}$$

$$\eta = \frac{P_s}{P} \times 100\%$$

$$= \frac{31900}{48000} \times 100\%$$

$$\Rightarrow \eta = 66.5\%$$

Design Output Summary

Rivet Hole d = 19 mm | Pitch p = 60 mm | Cover Strap t1 = 6.25 mm | Failure Mode: Shearing | Efficiency eta = 66.5%

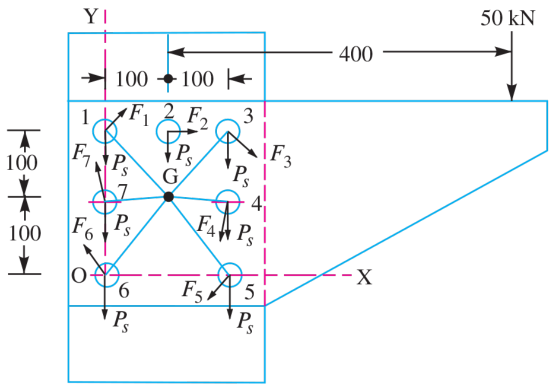


Fig. 9.25

Fig. 9.25: Single Riveted Double Strap Butt Joint

SECTION 7: BOILER LONGITUDINAL DOUBLE RIVETED BUTT JOINT

System Parameters: Boiler Shell Diameter $D = 1.5\ \text{m} = 1500\ \text{mm}$, Steam Pressure $P = 0.95\text{N/mm}^2$, Target Efficiency $\eta_l = 75\%$, Tensile Stress $\sigma_t = 90\ \text{MPa}$, Crushing Stress $\sigma_c = 140\ \text{MPa}$, Shear Stress $\tau = 56\ \text{MPa}$, Double Shear Factor = 1.875, Rivet Rows $n = 2$.

Design Protocol: Calculate shell thickness t, size rivet hole diameter d, state symbolic tearing (P_t) and shearing (P_s) capacities to solve pitch p, limit by I.B.R. max pitch ($p \le p_{\max}$), calculate row pitch p_b, strap thickness t_1, and margin m, evaluate failure capacities (P_t, P_s, P_c), and verify joint efficiency η.

1. Boiler Shell Plate Thickness (t)
Thin cylinder formula with 1 mm allowance

$$t = \frac{P \cdot D}{2\sigma_t \cdot \eta_l} + 1\ \text{mm}$$
$$= \frac{0.95 \times 1500}{2 \times 90 \times 0.75} + 1$$
$$= \frac{1425}{135} + 1 = 10.55 + 1 = 11.55\ \text{mm}$$
$$\Rightarrow \textbf{Shell Thickness}\ \ \boldsymbol{t = 12\ \text{mm}}$$

2. Diameter of Rivet Hole (d)
Unwin's formula for $t = 12\ \text{mm} > 8\ \text{mm}$ and IS standard size

$$d = 6\sqrt{t}$$
$$= 6\sqrt{12} = 20.78\ \text{mm}$$
$$\Rightarrow \textbf{Standard Hole Diameter}\ \ \boldsymbol{d = 21\ \text{mm}}$$
$$\Rightarrow \textbf{Nominal Rivet Diameter}\ \ \boldsymbol{d_0 = 20\ \text{mm}}$$

3. Tearing Resistance of Plate (P_t)
Net tensile area capacity per pitch length

$$P_t = (p - d)t \cdot \sigma_t$$
$$= (p - 21) \times 12 \times 90$$
$$= 1080(p - 21)\ \text{N}$$

4. Shearing Resistance of Rivets (P_s)
Double shear capacity for 2 rivets per pitch (1.875 × single shear)

$$P_s = n \cdot 1.875 \cdot \frac{\pi}{4} d^2 \cdot \tau$$
$$= 2 \times 1.875 \times \frac{\pi}{4}(21)^2 \times 56$$
$$= 3.75 \times 346.36 \times 56$$
$$= \mathbf{72745\ N}$$

5. Calculated Pitch & I.B.R. Max Pitch Limit ($p, p_{\max}$)
Equating $P_t = P_s$ and enforcing I.B.R. maximum pitch limit

$$1080(p - 21) = 72745\ \text{N}$$
$$p - 21 = \frac{72745}{1080} = 67.35 \Rightarrow p = 88.35\ \text{mm}$$
$$p_{\max} = C \cdot t + 41.28\ \text{mm}$$
$$= 3.5(12) + 41.28 = 83.28\ \text{mm}$$
$$\Rightarrow \textbf{Adopt Pitch}\ \ \boldsymbol{p = p_{\max} = 84\ \text{mm}}$$

6. Distance Between Rows of Rivets (p_b)
Transverse row pitch for zig-zag riveting

$$p_b = 0.33p + 0.67d$$
$$= 0.33(84) + 0.67(21)$$
$$= 27.72 + 14.07 = 41.79 \text{ mm}$$
$$\Rightarrow \textbf{Adopt Row Pitch } \ p_b = \textbf{42 mm}$$

---

7. Thickness of Cover Plates (t_1)
Standard proportion for equal width double cover straps

$$t_1 = 0.625t$$
$$= 0.625(12)$$
$$\Rightarrow \textbf{Adopt Strap Thickness } \ t_1 = \textbf{7.5 mm}$$

---

8. Margin (m)
Distance from rivet hole center to plate edge

$$m = 1.5d$$
$$= 1.5(21) = 31.5 \text{ mm}$$
$$\Rightarrow \textbf{Adopt Margin } \ m = \textbf{32 mm}$$

---

9. Governing Failure Mode & Joint Efficiency Verification (η)
Tearing capacity P_t, shearing P_s, crushing P_c, solid plate strength P, and efficiency

$$P_t = (84 - 21)(12)(90) = 68040 \text{ N (Governing)}$$
$$P_s = 72745 \text{ N}$$
$$P_c = n \cdot d \cdot t \cdot \sigma_c = 2(21)(12)(140) = 70560 \text{ N}$$
$$P = p \cdot t \cdot \sigma_t = 84(12)(90) = 90720 \text{ N}$$
$$\eta = \frac{P_t}{P} \times 100\%$$
$$= \frac{68040}{90720} \times 100\%$$
$$\Rightarrow \eta = 75.0\% \quad \text{(Verified matching target efficiency 75\%}$$

Design Output Summary

Shell t = 12 mm | Hole d = 21 mm | Pitch p = 84 mm | Row Pitch pb = 42 mm | Strap t1 = 7.5 mm | Margin m = 32 mm | Efficiency eta = 75%

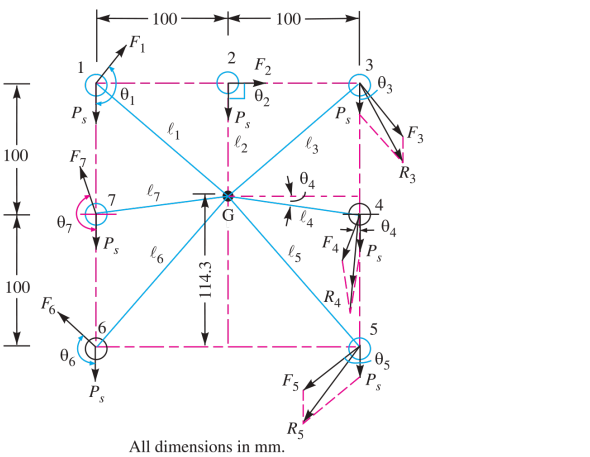


Fig. 9.26

Fig. 9.26: Boiler Longitudinal Double Riveted Double Strap Butt Joint

SECTION 8: PRESSURE VESSEL DOUBLE STRAP BUTT JOINT

System Parameters: Vessel Diameter $D = 1\text{ m} = 1000\text{ mm}$, Internal Pressure $P = 2.75\text{N/mm}^2$, Target Efficiency $\eta_l = 79\%$, Tensile Stress $\sigma_t = 88\text{ MPa}$, Shear Stress $\tau = 64\text{ MPa}$, Double Shear Factor = 1.8, Zig-zag outer pitch $p = 2p_{\text{inner}}$, $n = 3$.

Design Protocol: Calculate shell thickness t, size rivet hole diameter d, state symbolic tearing (P_t) and shearing (P_s) capacities to solve outer pitch p, limit by I.B.R. max pitch ($p \le p_{\text{max}}$), compute inner pitch p_{inner}, row pitch p_b, strap thickness t_1, margin m, and evaluate joint efficiency η.

---

1. Shell Plate Thickness (t)
Thin cylinder formula with 1 mm corrosion/manufacturing allowance

$$t = \frac{P \cdot D}{2\sigma_t \cdot \eta_l} + 1\text{ mm}$$
$$= \frac{2.75 \times 1000}{2 \times 88 \times 0.79} + 1$$
$$= \frac{2750}{139.04} + 1 = 19.78 + 1 = 20.78\text{ mm}$$
$$\Rightarrow \textbf{Shell Thickness}\;\; \boldsymbol{t = 21\text{ mm}}$$

2. Diameter of Rivet Hole (d)
Unwin's formula for $t = 21\text{ mm} > 8\text{ mm}$ and IS standard size

$$d = 6\sqrt{t}$$
$$= 6\sqrt{21} = 27.5\text{ mm}$$
$$\Rightarrow \textbf{Standard Hole Diameter}\;\; \boldsymbol{d = 28.5\text{ mm}}$$
$$\Rightarrow \textbf{Nominal Rivet Diameter}\;\; \boldsymbol{d_0 = 27\text{ mm}}$$

3. Tearing Resistance of Plate (P_t)
Net tensile area capacity per pitch length in outer row

$$P_t = (p - d)t \cdot \sigma_t$$
$$= (p - 28.5) \times 21 \times 88$$
$$= 1848(p - 28.5)\text{ N}$$

4. Shearing Resistance of Rivets (P_s)
Shear capacity for 3 rivets in double shear (1.8 × single shear)

$$P_s = n \cdot 1.8 \cdot \frac{\pi}{4} d^2 \cdot \tau$$
$$= 3 \times 1.8 \times \frac{\pi}{4}(28.5)^2 \times 64$$
$$= 5.4 \times 637.9 \times 64$$
$$= \mathbf{220500\text{ N}}$$

5. Outer & Inner Pitches ($p_{\text{outer}}, p_{\text{inner}}$)
Equating $P_t = P_s$ and enforcing I.B.R. maximum pitch limit

$$1848(p - 28.5) = 220500\text{ N}$$
$$p - 28.5 = \frac{220500}{1848} = 119.3 \Rightarrow p = 147.8\text{ mm}$$
$$p_{\text{max}} = C \cdot t + 41.28\text{ mm}$$
$$= 4.63(21) + 41.28 = 138.51\text{ mm}$$
$$\Rightarrow \textbf{Adopt Outer Pitch}\;\; \boldsymbol{p_{\text{outer}} = p_{\text{max}} = 14}$$
$$\Rightarrow \textbf{Adopt Inner Pitch}\;\; \boldsymbol{p_{\text{inner}} = \frac{p}{2} = \frac{140}{2} =}$$

6. Distance Between Rows of Rivets (p_b)

I.B.R. row spacing for outer pitch double inner pitch zig-zag butt joint

$$p_b = 0.2p + 1.15d$$

$$= 0.2(140) + 1.15(28.5)$$

$$= 28 + 32.775 = 60.775 \text{ mm}$$

$$\Rightarrow \textbf{Adopt Row Pitch } \; \boldsymbol{p_b = 61} \textbf{ mm}$$

7. Thickness of Butt Straps (t_1)

Equal width double butt straps ratio formula

$$t_1 = 0.625t \cdot \left(\frac{p-d}{p-2d}\right)$$

$$= 0.625(21) \times \left(\frac{140 - 28.5}{140 - 2(28.5)}\right)$$

$$= 13.125 \times \left(\frac{111.5}{83}\right) = 17.63 \text{ mm}$$

$$\Rightarrow \textbf{Adopt Strap Thickness } \; \boldsymbol{t_1 = 18} \textbf{ mm}$$

8. Margin (m)

Distance from rivet hole center to plate edge

$$m = 1.5d$$

$$= 1.5(28.5) = 42.75 \text{ mm}$$

$$\Rightarrow \textbf{Adopt Margin } \; \boldsymbol{m = 43} \textbf{ mm}$$

9. Governing Failure Mode & Joint Efficiency Verification (η)

Tearing capacity P_t, solid plate strength P, and efficiency

$$P_t = (p - d)t \cdot \sigma_t = (140 - 28.5)(21)(88) = 20605$$

$$P_s = 220500 \text{ N}$$

$$P = p \cdot t \cdot \sigma_t = 140(21)(88) = 258720 \text{ N}$$

$$\eta = \frac{P_t}{P} \times 100\%$$

$$= \frac{206050}{258720} \times 100\%$$

$$\Rightarrow \boldsymbol{\eta = 79.6\%} \quad (\textbf{Satisfactory}, \geq \textbf{79\%} \;)$$

Design Output Summary

Shell t = 21 mm | Hole d = 28.5 mm | Outer Pitch = 140 mm | Inner Pitch = 70 mm | Row Pitch pb = 61 mm | Strap t1 = 18 mm | Margin m = 43 mm | Efficiency eta = 79.6%

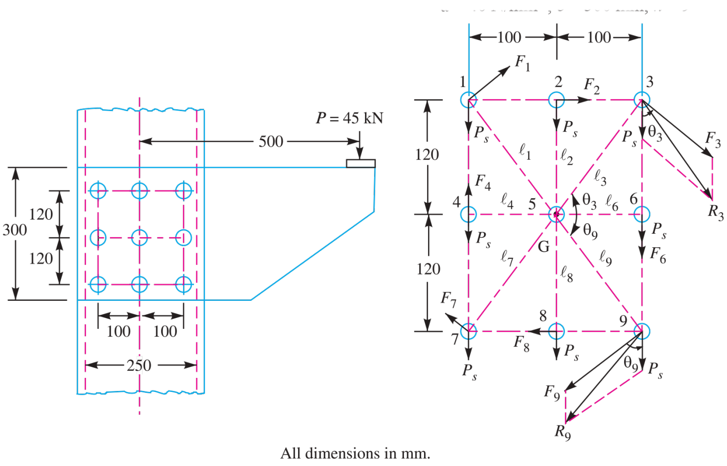


Fig. 9.27

Fig. 9.28

Fig. 9.27 & 9.28: Pressure Vessel Double Strap Butt Joint

SECTION 9: BOILER LONGITUDINAL SEAM DESIGN

System Parameters: Boiler Diameter $D = 1.25\ \mathrm{m} = 1250\ \mathrm{mm}$, Pressure $P = 2.5\mathrm{N/mm}^2$, Ultimate Tensile Stress $\sigma_{\mathrm{tu}} = 420\ \mathrm{MPa}$, Ultimate Crushing Stress $\sigma_{\mathrm{cu}} = 650\ \mathrm{MPa}$, Ultimate Shear Stress $\tau_{\mathrm{u}} = 300\ \mathrm{MPa}$, Target Efficiency $\eta_l = 80\%$, Factor of Safety $\mathrm{FOS} = 5$, $n = 5$ rivets per pitch (4 double, 1 single shear).

Design Protocol: Calculate allowable working stresses (σ_t, σ_c, τ), size plate thickness t and rivet hole diameter d, state symbolic tearing (P_t) and shearing (P_s) capacities to solve pitch p, limit by I.B.R. max pitch ($p \le p_{\mathrm{max}}$), compute inner pitch p', row pitches ($p_{\mathrm{b1}}, p_{\mathrm{b2}}$), wide ($t_1$) and narrow ($t_2$) strap thicknesses, margin m, and evaluate governing combined failure mode ($P_{\mathrm{t2}} + P_{\mathrm{s1}}$) to verify joint efficiency η.

---

1. Allowable Working Stresses
Factor of safety FOS = 5 applied to ultimate material strengths

$$\sigma_t = \frac{\sigma_{\mathrm{tu}}}{\mathrm{FOS}} = \frac{420}{5} = 84\mathrm{N/mm}^2$$
$$\sigma_c = \frac{\sigma_{\mathrm{cu}}}{\mathrm{FOS}} = \frac{650}{5} = 130\mathrm{N/mm}^2$$
$$\tau = \frac{\tau_{\mathrm{u}}}{\mathrm{FOS}} = \frac{300}{5} = 60\mathrm{N/mm}^2$$

2. Boiler Shell Plate Thickness (t)
Thin cylinder formula with 1 mm corrosion/manufacturing allowance

$$t = \frac{P \cdot D}{2\sigma_t \cdot \eta_l} + 1\ \mathrm{mm}$$
$$= \frac{2.5 \times 1250}{2 \times 84 \times 0.80} + 1$$
$$= \frac{3125}{134.4} + 1 = 23.25 + 1$$
$$= 24.25\ \mathrm{mm}$$
$$\Rightarrow \textbf{Plate Thickness}\ \ \boldsymbol{t = 25\ \mathrm{mm}}$$

3. Diameter of Rivet Hole (d)
Unwin's formula for $t = 25\ \mathrm{mm} > 8\ \mathrm{mm}$ and IS standard size

$$d = 6\sqrt{t}$$
$$= 6\sqrt{25} = 30\ \mathrm{mm}$$
$$\Rightarrow \textbf{Standard Hole Diameter}\ \ \boldsymbol{d = 31.5\ \mathrm{mm}}$$
$$\Rightarrow \textbf{Nominal Rivet Diameter}\ \ \boldsymbol{d_0 = 30\ \mathrm{mm}}$$

4. Tearing Resistance of Plate at Outer Row (P_t)
Net tensile area capacity per pitch length

$$P_t = (p - d)t \cdot \sigma_t$$
$$= (p - 31.5) \times 25 \times 84$$
$$= 2100(p - 31.5)\ \mathrm{N}$$

5. Shearing Resistance of Rivets (P_s)
Combined shear capacity for 4 rivets in double shear + 1 rivet in single shear

$$P_s = (4 \cdot 1.875 + 1) \cdot \frac{\pi}{4} d^2 \cdot \tau$$
$$= 8.5 \times \frac{\pi}{4}(31.5)^2 \times 60$$
$$= 8.5 \times 779.3 \times 60$$
$$= \mathbf{397500\ N}$$

6. Calculated Pitch & I.B.R. Max Pitch Limit ($p, p_{\max}$)

Equating $P_t = P_s$ and enforcing I.B.R. maximum pitch limit

$$2100(p - 31.5) = 397500 \text{ N}$$

$$p - 31.5 = \frac{397500}{2100} = 189.3 \Rightarrow p = 220.8 \text{ mm}$$

$$p_{\max} = C \cdot t + 41.28 \text{ mm}$$

$$= 6 \times 25 + 41.28 = 191.28 \text{ mm}$$

$$\Rightarrow \textbf{Adopt Outer Pitch } \; \boldsymbol{p = p_{\max}} = \textbf{196 m}$$

$$\Rightarrow \textbf{Adopt Inner Pitch } \; \boldsymbol{p' = \frac{p}{2} = \frac{196}{2} = 9}$$

---

7. Distance Between Rows of Rivets (p_{b1}, p_{b2})

I.B.R. row spacing for outer-to-next row and inner zig-zag rows

$$p_{\text{b1}} = 0.2p + 1.15d$$

$$= 0.2(196) + 1.15(31.5) = 39.2 + 36.225 = 75.425 \text{ mm}$$

$$\Rightarrow \textbf{Outer Row Spacing } \; \boldsymbol{p_{\text{b1}}} = \textbf{76 mm}$$

$$p_{\text{b2}} = 0.165p + 0.67d$$

$$= 0.165(196) + 0.67(31.5) = 32.34 + 21.105 = 53.445 \text{ mm}$$

$$\Rightarrow \textbf{Inner Row Spacing } \; \boldsymbol{p_{\text{b2}}} = \textbf{54 mm}$$

---

8. Butt Strap Thicknesses & Margin (t_1, t_2, m)

Inside wide strap, outside narrow strap, and margin

$$t_1 = 0.75t = 0.75(25) = 18.75 \text{ mm} \Rightarrow \textbf{Wide Stra}$$

$$t_2 = 0.625t = 0.625(25) = 15.625 \text{ mm} \Rightarrow \textbf{Narrow}$$

$$m = 1.5d = 1.5(31.5) = 47.25 \text{ mm} \Rightarrow \textbf{Margin } \; \boldsymbol{m}$$

---

9. Individual Failure Mode Capacities

Tearing (P_t), shearing (P_s), and crushing (P_c) capacities for $p = 196$ mm

$$P_{\text{t1}} = (p - d)t \cdot \sigma_t = (196 - 31.5)(25)(84) = 34545$$

$$P_s = 8.5 \cdot \frac{\pi}{4} d^2 \cdot \tau = 397500 \text{ N}$$

$$P_c = n \cdot d \cdot t \cdot \sigma_c = 5 \times 31.5 \times 25 \times 130 = 511875$$

---

10. Combined Row 2 Tearing & Outer Rivet Shear Capacity

Tearing of plate at second row combined with shearing 1 outer rivet

$$P_{\text{t2}} + P_{\text{s1}} = (p - 2d)t \cdot \sigma_t + \frac{\pi}{4} d^2 \cdot \tau$$

$$= (196 - 2 \times 31.5) \times 25 \times 84 + \frac{\pi}{4}(31.5)^2 \times 60$$

$$= (133)(25)(84) + 47815$$

$$= 278250 + 47815$$

$$\Rightarrow \textbf{Governing Least Strength } \; \boldsymbol{P_u} = \textbf{326065 N}$$

---

11. Solid Plate Strength & Joint Efficiency (P, η)

Un-punched solid plate strength and overall joint efficiency

$$P = p \cdot t \cdot \sigma_t$$

$$= 196 \times 25 \times 84$$

$$= 411600 \text{ N}$$

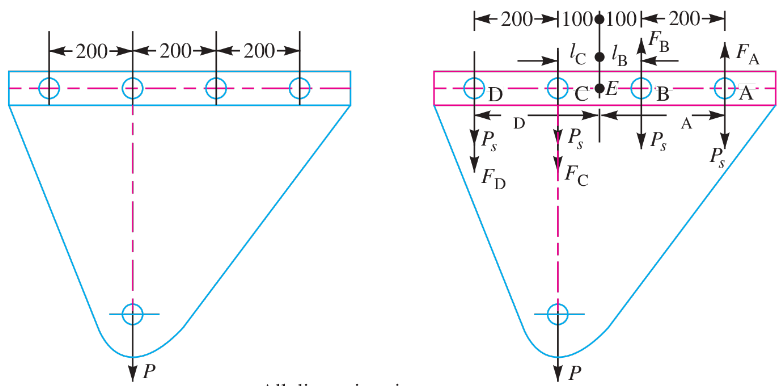
$$\eta = \frac{P_u}{P} \times 100\%$$

$$= \frac{326065}{411600} \times 100\%$$

$$\Rightarrow \boldsymbol{\eta = 79.2\%} \quad \textbf{(Satisfactory, nearly equal to target 80\%}$$

Design Output Summary

Shell t = 25 mm | Hole d = 31.5 mm | Outer Pitch = 196 mm | Inner Pitch = 98 mm | Row Spacings = 76 mm & 54 mm | Straps t1 = 20 mm, t2 = 16 mm | Margin m = 47.5 mm | Governing Strength = 326065 N | Efficiency eta = 79.2%



All dimensions in mm.

Fig. 9.29

Fig. 9.30

Fig. 9.29 & 9.30: Boiler Longitudinal Seam Triple Riveted Butt Joint

SECTION 10: BOILER LONGITUDINAL & CIRCUMFERENTIAL JOINTS

System Parameters: Boiler Diameter $D = 1.6\ \text{m} = 1600\ \text{mm}$, Pressure $p_i = 2.5\text{N/mm}^2$, Allowable Stresses: Tensile $\sigma_t = 75\ \text{MPa}$, Shear $\tau = 60\ \text{MPa}$, Crushing $\sigma_c = 125\ \text{MPa}$.

Design Protocol: Summarize longitudinal joint plate thickness t and rivet diameter d; design circumferential joint by calculating total steam thrust F and required rivet count N_c.

1. Longitudinal Joint Parameters
Plate thickness and rivet diameter selection

$t = 35\ \text{mm}$
$d = 35.5\ \text{mm}$

2. Total Steam End Thrust (F)
Axial burst force acting on boiler end cap

$$F = \frac{\pi}{4} D^2 \cdot p_i$$
$$= \frac{\pi}{4}(1600)^2 \times 2.5$$
$$= 5.026 \times 10^6\ \text{N}$$

3. Number of Circumferential Rivets (N_c)
Number of single shear rivets required to resist axial thrust

$$N_c = \frac{F}{\frac{\pi}{4} d^2 \cdot \tau}$$
$$= \frac{5.026 \times 10^6}{\frac{\pi}{4}(35.5)^2 \times 60}$$
$$= \frac{5.026 \times 10^6}{59390}$$
$$= 84.6$$
$$\Rightarrow \boldsymbol{N_c = 85\ \text{rivets}}$$

Design Output

Longitudinal: t = 35 mm, d = 35.5 mm | Circumferential Joint Thrust F = 5.026 MN | Rivet Count Nc = 85 rivets

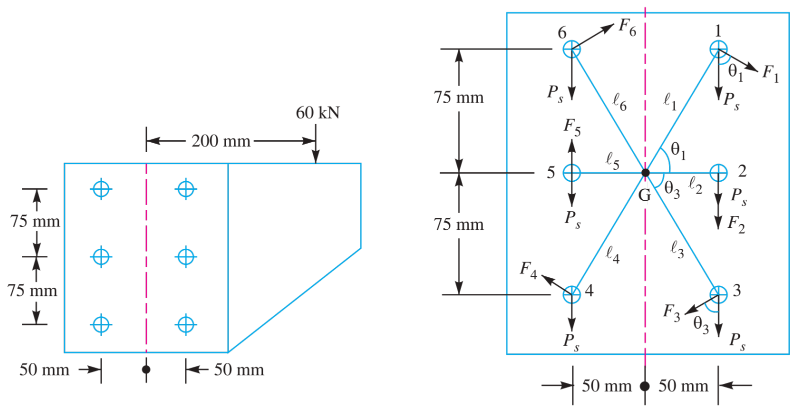


Fig. 9.31 Fig. 9.32

Fig. 9.31 & 9.32: Boiler Longitudinal and Circumferential Joints

SECTION 11: TIE ROD DOUBLE COVER BUTT JOINT

System Parameters: Tie Rod Cross-Section $b = 200$ mm, $t = 12.5$ mm, Allowable Stresses: Tensile $\sigma_t = 80$ MPa, Shear $\tau = 65$ MPa, Crushing $\sigma_c = 160$ MPa, Double Shear Factor = 1.875.

Design Protocol: Calculate rivet diameter d, evaluate maximum tensile load capacity P, compute single rivet double shear strength P_s, and determine required rivet count n.

1. Rivet Diameter (d)
Unwin's formula for $t = 12.5$ mm

$$d = 6\sqrt{t}$$
$$= 6\sqrt{12.5}$$
$$= 21.21 \text{ mm}$$
$$\Rightarrow \boldsymbol{d = 21.5 \text{ mm}}$$

2. Full Tensile Load (P)
Solid tie-rod tensile capacity

$$P = b \cdot t \cdot \sigma_t$$
$$= 200 \times 12.5 \times 80$$
$$= 200000 \text{ N}$$
$$= 200 \text{ kN}$$

3. Single Rivet Double Shear Capacity (P_s)
Shear strength per rivet in double cover butt joint

$$P_s = 1.875 \cdot \frac{\pi}{4} d^2 \cdot \tau$$
$$= 1.875 \times \frac{\pi}{4}(21.5)^2 \times 65$$
$$= 44250 \text{ N}$$
$$= 44.25 \text{ kN}$$

4. Required Rivet Count (n)
Total rivets needed per side of joint

$$n = \frac{P}{P_s}$$
$$= \frac{200}{44.25}$$
$$= 4.52$$
$$\Rightarrow \boldsymbol{n = 5 \text{ rivets}}$$

Design Output

Tie Rod Joint: Rivet Diameter d = 21.5 mm | Tensile Load P = 200 kN | Required Rivets n = 5 rivets

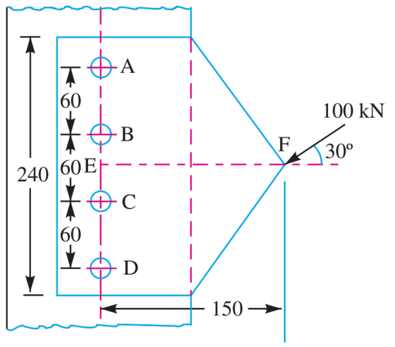


Fig. 9.33

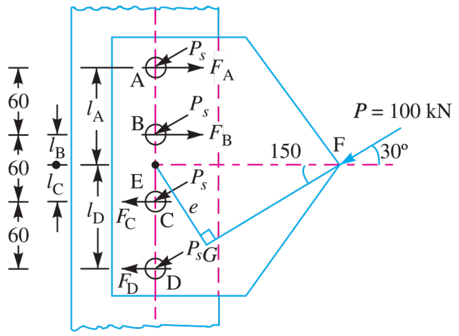


Fig. 9.34

Fig. 9.33 & 9.34: Tie Rod Double Cover Butt Joint

SECTION 12: BRIDGE TIE-BAR LOZENGE (DIAMOND) JOINT

System Parameters: Flat Bar Width $b = 350$ mm, Thickness $t = 20$ mm, Allowable Stresses: Tensile $\sigma_t = 90$ MPa, Shear $\tau = 60$ MPa, Crushing $\sigma_c = 150$ MPa, Double Shear Factor = 1.875.

Design Protocol: Calculate rivet diameter d, evaluate net load capacity P at outer row (1 rivet), determine single rivet double shear strength P_s, compute required number of rivets n, and arrange in diamond pattern.

---

1. Rivet Diameter (d)
Unwin's formula for $t = 20$ mm

$$d = 6\sqrt{t}$$
$$= 6\sqrt{20}$$
$$= 26.83 \text{ mm}$$
$$\Rightarrow \boldsymbol{d = 27} \textbf{ mm}$$

2. Net Load Capacity (P)
Tearing strength at critical Section 1-1 (1 rivet hole)

$$P = (b - d)t \cdot \sigma_t$$
$$= (350 - 27) \times 20 \times 90$$
$$= 323 \times 1800$$
$$= 581400 \text{ N}$$
$$= 581.4 \text{ kN}$$

3. Single Rivet Double Shear Capacity (P_s)
Shear resistance per rivet

$$P_s = 1.875 \cdot \frac{\pi}{4} d^2 \cdot \tau$$
$$= 1.875 \times \frac{\pi}{4}(27)^2 \times 60$$
$$= 64412 \text{ N}$$
$$= 64.4 \text{ kN}$$

4. Total Rivet Count (n)
Total rivets required for lozenge arrangement

$$n = \frac{P}{P_s}$$
$$= \frac{581.4}{64.4}$$
$$= 9.02$$
$$\Rightarrow \boldsymbol{n = 9} \textbf{ rivets}$$

5. Lozenge Rivet Layout
Symmetric diamond pattern from outer row to inner row

Arranged in Diamond Pattern: Row 1 (1 rivet), Row 2 (2 rivets), Row 3 (3 rivets), Row 4 (3 rivets)

Design Output

Bridge Lozenge Joint: d = 27 mm | Net Capacity P = 581.4 kN | Total Rivets n = 9 (Pattern: 1, 2, 3, 3)

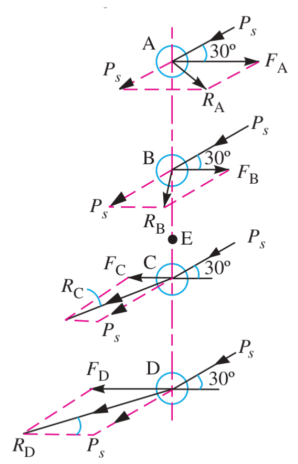


Fig. 9.35

Fig. 9.35: Bridge Tie-Bar Lozenge Joint

SECTION 13: FLAT TIE-BAR LAP JOINT (DIAMOND PATTERN)

System Parameters: Flat Bar Width $b = 200$ mm, Thickness $t = 10$ mm, Rivet Shank Diameter $d = 24$ mm, Rivet Hole Diameter $d_h = 25.5$ mm, Allowable Stresses: Tensile $\sigma_t = 112$ MPa, Shear $\tau = 84$ MPa, Crushing $\sigma_c = 200$ MPa.

Design Protocol: Arrange 4 rivets in diamond pattern (1, 1, 2); evaluate net tearing strength at outer single-rivet row, and compute joint efficiency.

---

1. Rivet Layout & Pattern
Specified rivet arrangement on flat tie-bar

n = 4 rivets arranged in Diamond Pattern (Row 1: 1, Row 2: 1, Row 3: 2)

2. Net Tearing Strength (P_t)
Tearing capacity at Section 1-1 with hole diameter $d_h = 25.5$ mm

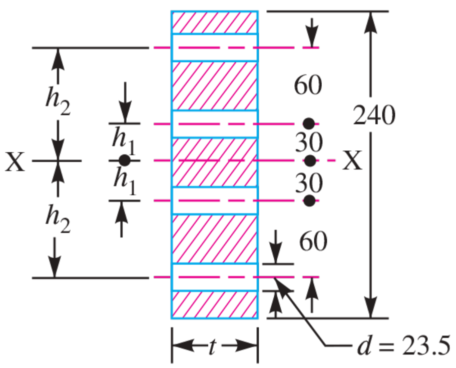
$$P_t = (b - d_h)t \cdot \sigma_t$$
$$= (200 - 25.5) \times 10 \times 112$$
$$= 174.5 \times 1120$$
$$= 195440 \text{ N}$$
$$= 195.44 \text{ kN}$$

3. Solid Strength (P) & Efficiency (η)
Efficiency based on net tearing capacity

$$P = b \cdot t \cdot \sigma_t$$
$$= 200 \times 10 \times 112$$
$$= 224000 \text{ N}$$
$$\eta = \frac{P_t}{P} \times 100\%$$
$$= \frac{195440}{224000} \times 100\%$$
$$\Rightarrow \boldsymbol{\eta = 87.3\%}$$

Design Output

Flat Lap Joint (Diamond Pattern): Net Tearing Strength Pt = 195.44 kN | Joint Efficiency eta = 87.3%



All dimensions in mm.

Fig. 9.36

Fig. 9.36: Flat Tie-Bar Lap Joint (Diamond Pattern)

SECTION 14: ECCENTRIC BRACKET WITH VERTICAL RIVET LINE

System Parameters: Plate Thickness $t = 25\ \mathrm{mm}$, Eccentric Load $P = 50\ \mathrm{kN} = 50000\ \mathrm{N}$, Eccentricity $e = 400\ \mathrm{mm}$, Rivet Spacing $C = 100\ \mathrm{mm}$, Number of Rivets $n = 4$ in vertical line, Allowable Shear Stress $\tau = 65\ \mathrm{MPa}$, Allowable Crushing Stress $\sigma_c = 120\ \mathrm{MPa}$.

Design Protocol: Calculate direct shear force F_{s1} per rivet; compute secondary shear force F_{s2} due to moment Pe; find max resultant shear force F_R, and solve for rivet diameter d.

---

1. Direct Shear Force (F_{s1})
Uniform load distribution across 4 rivets

$$F_{\mathrm{s1}} = \frac{P}{n}$$
$$= \frac{50000}{4}$$
$$= 12500\ \mathrm{N}$$
$$= 12.5\ \mathrm{kN}$$

---

2. Secondary Shear Force (F_{s2})
Torsional shear force on outermost rivet ($r_{\max} = 150\ \mathrm{mm}$)

$$\sum r_i^2 = 2(150^2 + 50^2)$$
$$= 2(22500 + 2500)$$
$$= 50000 \mathrm{mm}^2$$
$$F_{\mathrm{s2}} = \frac{P \cdot e \cdot r_{\max}}{\sum} r_i^2$$
$$= \frac{50000 \times 400 \times 150}{50000}$$
$$= 60000\ \mathrm{N}$$
$$= 60\ \mathrm{kN}$$

---

3. Maximum Resultant Shear Force (F_R)
Direct plus secondary shear force (co-linear theta = 0 deg)

$$F_R = F_{\mathrm{s1}} + F_{\mathrm{s2}}$$
$$= 12.5 + 60$$
$$= 72.5\ \mathrm{kN}$$
$$= 72500\ \mathrm{N}$$

---

4. Rivet Diameter (d)
Equating F_R to rivet single shear strength

$$F_R = \frac{\pi}{4} d^2 \cdot \tau$$
$$72500 = \frac{\pi}{4} d^2 (65)$$
$$d^2 = \frac{4 \times 72500}{\pi \times 65}$$
$$= \frac{290000}{204.2}$$
$$= 1420.2$$
$$d = 37.68\ \mathrm{mm}$$
$$\Rightarrow \boldsymbol{d = 38\ \mathrm{mm}}$$

Design Output

Eccentric Bracket: Direct Shear Fs1 = 12.5 kN | Secondary Shear Fs2 = 60 kN | Resultant FR = 72.5 kN | d = 38 mm

SECTION 15: ECCENTRICALLY LOADED BRACKET JOINT

System Parameters: Eccentric Load $P = 45\ \text{kN}$, Allowable Shear Stress $\tau \leq 40\ \text{MPa}$.

Design Protocol: Evaluate direct and secondary shear forces to determine maximum resultant shear load $F_R = 15.2\ \text{kN}$; size rivet diameter d.

1. Maximum Resultant Shear Force (F_R) Critical rivet resultant load from direct and secondary shear	$F_R = 15.2\ \text{kN}$ $= 15200\ \text{N}$
2. Rivet Diameter (d) Single shear area calculation for allowable stress $\tau = 40\ \text{MPa}$	$d = \sqrt{\dfrac{4F_R}{\pi\tau}}$ $= \sqrt{\dfrac{4 \times 15200}{\pi \times 40}}$ $= \sqrt{\dfrac{60800}{125.66}}$ $= \sqrt{483.8}$ $= 22.00\ \text{mm}$ $\Rightarrow \boldsymbol{d = 22\ \text{mm}}$
Design Output	**Eccentric Bracket (45 kN Load): Critical Resultant FR = 15.2 kN | Rivet Diameter d = 22 mm**

SECTION 16: MAXIMUM SAFE ECCENTRIC LOAD CAPACITY

System Parameters: Number of Rivets $n = 4$, Rivet Diameter $d = 20\ \text{mm}$, Working Shear Stress $\tau = 100\ \text{MPa}$.

Design Protocol: Determine maximum single rivet shear load capacity $F_{\text{R,max}}$; express critical resultant force as a function of external load P ($F_R = 0.966P$); solve maximum safe load P.

1. Single Rivet Shear Capacity ($F_{\text{R,max}}$) Maximum permissible shear force per rivet	$F_{\text{R,max}} = \frac{\pi}{4}d^2 \cdot \tau$ $= \frac{\pi}{4}(20)^2 \times 100$ $= 31416\ \text{N}$ $= 31.416\ \text{kN}$
2. Resultant Load Relation Critical rivet resultant load expressed in terms of load P	$F_R = 0.966P$
3. Maximum Safe Load (P) Equating F_R to single rivet shear capacity	$P = \frac{F_{\text{R,max}}}{0.966}$ $= \frac{31.416}{0.966}$ $= 32.52\ \text{kN}$ $\Rightarrow \boldsymbol{P = 32.5\ \text{kN}}$
Design Output	**Eccentric Load Capacity: Single Rivet Capacity FR,max = 31.416 kN | Maximum Safe Load P = 32.5 kN**

SECTION 17: ECCENTRIC COLUMN BRACKET JOINT

System Parameters: Number of Rivets $n = 6$, Eccentric Load $P = 60 \text{ kN} = 60000 \text{ N}$, Eccentricity $e = 200 \text{ mm}$, Allowable Shear Stress $\tau \leq 150 \text{ MPa}$.

Design Protocol: Calculate primary direct shear F_{s1} and secondary shear F_{s2}; vectorially combine to get critical resultant shear force F_R; solve rivet diameter d.

---

1. Primary & Secondary Shear Forces
Direct load per rivet and torsional secondary load

$$\begin{aligned} F_{\text{s1}} &= \frac{P}{n} \\ &= \frac{60}{6} \\ &= 10 \text{ kN} \\ F_{\text{s2}} &= \frac{P \cdot e \cdot r_{\text{max}}}{\sum} r_i^2 \\ &= 41.5 \text{ kN} \end{aligned}$$

2. Maximum Resultant Shear Force (F_R)
Vector addition of primary and secondary shear

$$\begin{aligned} F_R &= 51.5 \text{ kN} \\ &= 51500 \text{ N} \end{aligned}$$

3. Rivet Diameter (d)
Equating F_R to single shear capacity

$$\begin{aligned} d &= \sqrt{\frac{4F_R}{\pi\tau}} \\ &= \sqrt{\frac{4 \times 51500}{\pi \times 150}} \\ &= \sqrt{\frac{206000}{471.24}} \\ &= \sqrt{437.1} \\ &= 20.91 \text{ mm} \\ &\Rightarrow \boldsymbol{d = 21 \text{ mm}} \end{aligned}$$

Design Output

6-Rivet Column Bracket: Fs1 = 10 kN | Fs2 = 41.5 kN | Resultant FR = 51.5 kN | Rivet Diameter d = 21 mm

SECTION 18: ECCENTRICALLY LOADED INCLINED BRACKET JOINT

System Parameters: Number of Rivets $n = 4$ (A, B, C, D, vertical line, pitch 60 mm), Inclined Load $P = 100\ \mathrm{kN} = 100000\ \mathrm{N}$ at 30^{deg} to horizontal, Eccentricity $e = 150\ \mathrm{mm}$, Allowable Shear Stress $\tau = 80\ \mathrm{MPa}$, Allowable Bending Stress $\sigma_b = 125\ \mathrm{MPa}$, Plate Width $b = 240\ \mathrm{mm}$.

Design Protocol: Resolve load into vertical and horizontal components (P_V, P_H); calculate max resultant shear force F_R on critical rivet D to size diameter d; determine bending moment M on bracket plate to calculate plate thickness t.

---

1. Vertical & Horizontal Load Components
Resolving inclined 100 kN load at 30 degrees

$$
\begin{aligned}
P_V &= P \sin\left(30^{\mathrm{deg}}\right) \\
&= 100 \sin\left(30^{\mathrm{deg}}\right) \\
&= 50\ \mathrm{kN} \\
P_H &= P \cos\left(30^{\mathrm{deg}}\right) \\
&= 100 \cos\left(30^{\mathrm{deg}}\right) \\
&= 86.6\ \mathrm{kN}
\end{aligned}
$$

2. Critical Resultant Shear (F_R) & Rivet Diameter (d)
Resultant force on rivet D and single shear sizing

$$
\begin{aligned}
F_R &= 42.4\ \mathrm{kN} \\
&= 42400\ \mathrm{N} \\
d &= \sqrt{\frac{4F_R}{\pi\tau}} \\
&= \sqrt{\frac{4 \times 42400}{\pi \times 80}} \\
&= \sqrt{\frac{169600}{251.33}} \\
&= \sqrt{674.8} \\
&= 25.98\ \mathrm{mm} \\
&\Rightarrow \boldsymbol{d = 26\ \mathrm{mm}}
\end{aligned}
$$

3. Bending Moment (M) & Bracket Thickness (t)
Bending stress analysis of bracket plate

$$
\begin{aligned}
M &= P_H \cdot e \\
&= 86.6 \times 10^3 \times 150 \\
&= 1.30 \times 10^7\ \mathrm{N \cdot mm} \\
\sigma_b &= \frac{6M}{t \cdot b^2} \\
125 &= \frac{6 \times 1.30 \times 10^7}{t \times (240)^2} \\
&= \frac{7.80 \times 10^7}{57600t} \\
&= \frac{1354.17}{t} \\
t &= \frac{1354.17}{125} \\
&= 10.83\ \mathrm{mm} \\
&\Rightarrow \boldsymbol{t = 14\ \mathrm{mm}}
\end{aligned}
$$

Design Output	**Inclined Bracket Joint: PV = 50 kN, PH = 86.6 kN | Critical FR = 42.4 kN | d = 26 mm | Plate Thickness t = 14 mm**